

POWERFACTORY

PowerFactory 2024

Technical Reference

Voltage Protection

RelUlim, TypUlim

PF2024

POWER SYSTEM SOLUTIONS

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1 General Description

The Voltage Protection Model is a function for detecting :

- overvoltages,
- undervoltages and
- vector jumps.

The Model can be configured with the Voltage Protection Type (*TypeUlim*) and the Voltage Protection Element (*RelUlim*).

2 Setting the model parameters

2.1 Voltage Protection Type Input (*TypUlim*)

The Parameter IEC Symbol (*t:sfiec*) and ANSI Symbol (*t:sfansi*) allow the insertion of standard symbols for documentation purposes. A list of possible symbols is available, but any other symbol can also be entered.

With the parameter Function (*t:ifunc*), the user can select one of the following functions:

- Undervoltage
- Overvoltage
- Vector Shift

If the selected function is “Undervoltage” or “Overvoltage”, the range and unit for the voltage setting can be entered in the voltage field (*t:rUset* and *t:iunitu*). Additionally the range and unit for the Time Delay can be entered (*t:rTdel* and *t:iunitt*). If the selected function is “Vector Shift”, only one range for the delta angle in degrees can be entered (*t:rUang*). The “Vector Shift” does not require a time delay.

The number and the type of the input quantities processed can be selectet using the parameter Type (*t:itype*). The models for the different types are shown in section 3.

For all types except “Voltage (6ph)”, the reference voltage can be selected between “Unom” and “Unom / sqrt(3)” with the selection list based on (*t:puBaseV*).

2.2 Voltage Protection Element Input (*RelUlim*)

For the function “Undervoltage” or “Overvoltage” the voltage threshold and the time delay can be set using the Voltage (*e:Uset*) and the Time Delay (*c:Tdel*). If the Function “Vector Shift”, the angular threshold can be set using the Delta Angle (*e:Uang*).

If the measurement type (*t:ityp*) is “Voltage (6ph)”, the desired input voltages (phase-phase or phase-earth) can be selected. The leading unit of the pickup voltage is the specification of the type *TypUlim*. The values of the other units are calculated with the nominal voltage and the transmission ratio.

$$U_{p.u.} = \frac{U_{sec.V}}{U_{nom}}$$

$$U_{sec.V} = U_{p.u.} \cdot U_{nom} \quad (1)$$

$$t_{VT} = \frac{U_{pri.V}}{U_{sec.V}}$$

Where is:

$U_{p.u.}$ is the Voltage in [p.u.].

U_{nom} is the nominal Voltage in [V].

$U_{pri.V}$ is the rated primary voltage of the connected voltage transformer in [V].

$U_{sec.V}$ is the rated secondary voltage of the connected voltage transformer in [V].

t_{VT} is the transmission ratio of the connected voltage transformer.

3 Calculation Models

Depending on the combination of selected measurement Type and Function, four different calculation models are available. Please see the following sections for more information about the models and the relevant combinations for each. “Undervoltage” and “Overvoltage” are implemented in the following models:

- Single-Phase Under-/Overvoltage Model,
- 3-Phase Under-/Overvoltage Model or
- 6-Phase Under-/Overvoltage Model.

“Vector Shift” is only available as a 3-phase model.

3.1 Single-Phase Under-/Overvoltage Model

The Single-Phase Under-/Overvoltage Model is used for the following types:

- “Voltage (1ph)”
- “Earth Voltage (3*U0)”
- “Negative Sequence (U2)”
- “Positive Sequence (U1)”

3.1.1 Load Flow and Short-Circuit Calculation

Figure 3.1 shows the Single-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculation.

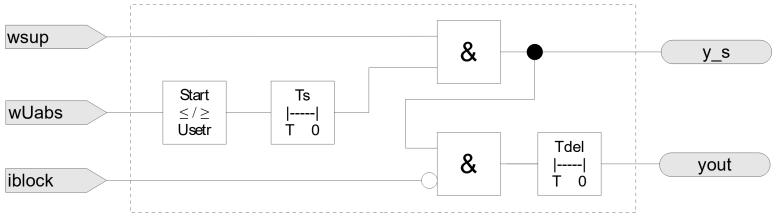


Figure 3.1: Logic scheme of the Single-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations

The detection of undervoltage and overvoltage is calculated as:

$$\begin{aligned} \text{Undervoltage:} \quad & \text{Start} = \begin{cases} \text{Ts}, & \text{for} \quad \text{wUabs} \leq \text{Usetr} \\ 9999.999, & \text{for} \quad \text{wUabs} > \text{Usetr} \end{cases} \\ \text{Overvoltage:} \quad & \text{Start} = \begin{cases} \text{Ts}, & \text{for} \quad \text{wUabs} \geq \text{Usetr} \\ 9999.999, & \text{for} \quad \text{wUabs} < \text{Usetr} \end{cases} \end{aligned} \tag{2}$$

Where:

Usetr is the Voltage Pickup.

Ts is the Pickup Time in [s].

The Starting Time (*s:y_s*) and the Tripping Time (*s:yout*) is calculated as:

$$\text{y_s} = \max(\text{wsup}, \text{Start})$$

$$\text{yout} = \begin{cases} (\text{y_s} + \text{Tdel}), & \text{for} \quad (\text{y_s} + \text{Tdel}) < \text{iblock} \\ 9999.999, & \text{for} \quad (\text{y_s} + \text{Tdel}) \geq \text{iblock} \end{cases} \tag{3}$$

Where:

Tdel is the Time Delay in [s].

Table 3.1: Input/output signals of the Single-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations

Signal	Description	Unit	Type
wsup	Supervising Signal	s	IN
wUabs	Voltage, Magnitude	sec.V	IN
iblock	Blocking Input	s	IN
y_s	Starting	s	OUT
yout	Tripping Time	s	OUT

Table 3.2: Parameters of the Single-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations

Parameter	Description	Unit
uabs	Voltage, Magnitude	p.u.
Uabs	Voltage, Magnitude	sec.V
Upabs	Voltage, Magnitude	pri.V

3.1.2 RMS and EMT Simulation

The Figure 3.2 shows the Single-Phase Under-/Overvoltage Model for RMS and EMT Simulations.

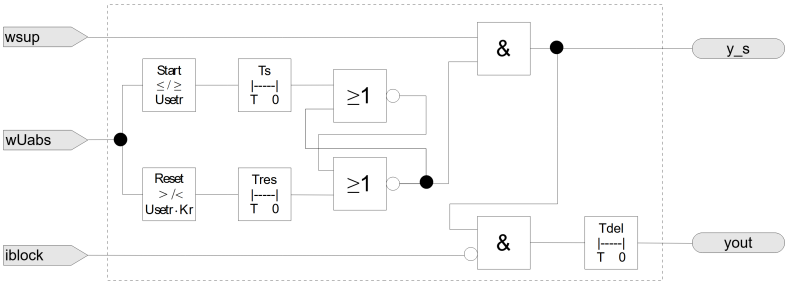


Figure 3.2: Logic scheme of the Single-Phase Under-/Overvoltage Model for RMS and EMT Simulations

The internal Start of undervoltage and overvoltage is calculated as:

$$\begin{aligned} \text{Undervoltage:} \quad \text{Start} &= \begin{cases} 1, & \text{for} \quad \text{wUabs} \leq \text{Usetr} \\ 0, & \text{for} \quad \text{wUabs} > \text{Usetr} \end{cases} \\ \text{Overvoltage:} \quad \text{Start} &= \begin{cases} 1, & \text{for} \quad \text{wUabs} \geq \text{Usetr} \\ 0, & \text{for} \quad \text{wUabs} < \text{Usetr} \end{cases} \end{aligned} \tag{4}$$

In the RMS and EMT Simulation the Start can fall back with internal Reset. It is calculated as:

$$\begin{aligned} \text{Undervoltage (with Kr} \leq 100\,\%\text{):} \quad \text{Reset} &= \begin{cases} 1, & \text{for} \quad \text{wUabs} \geq \text{Usetr} \cdot \text{Kr} \\ 0, & \text{for} \quad \text{wUabs} < \text{Usetr} \cdot \text{Kr} \end{cases} \\ \text{Overvoltage (with Kr} \geq 100\,\%\text{):} \quad \text{Reset} &= \begin{cases} 1, & \text{for} \quad \text{wUabs} \leq \text{Usetr} \cdot \text{Kr} \\ 0, & \text{for} \quad \text{wUabs} > \text{Usetr} \cdot \text{Kr} \end{cases} \end{aligned} \tag{5}$$

Where:

Usetr is the Voltage Pickup.

Kr is the Reset Ratio.

The starting signal (*s:y_s*) and the tripping signal (*s:yout*) is calculated as:

$$\begin{aligned} \text{y_s} &= \text{wsup} \wedge \neg\left(\neg\,\text{Start} \vee \text{Reset}\right) \\ \text{yout} &= \text{y_s} \wedge \neg\,\text{iblock} \end{aligned} \tag{6}$$

Table 3.3: Input/output signals of the Single-Phase Under-/Overvoltage Model for RMS and EMT Simulations

Signal	Description	Unit	Type
wsup	Supervising Signal		IN
wUabs	Voltage, Magnitude	sec.V	IN
iblock	Blocking Input		IN
y_s	Starting		OUT
yout	Tripping		OUT

Table 3.4: Parameters of the Single-Phase Under-/Overvoltage Model for RMS and EMT Simulations

Parameter	Description	Unit
Ttrip	Tripping Time	s
uabs	Voltage, Magnitude	p.u.
Uabs	Voltage, Magnitude	sec.V
Upabs	Voltage, Magnitude	pri.V

3.2 3-Phase Under-/Overvoltage Model

This model is used for the Type “Voltage 3ph”. The structure of the 3-phase model is similar to that of the single-phase model.

3.2.1 Load Flow and Short-Circuit Calculation

The Figure 3.3 shows the structure of the 3-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations.

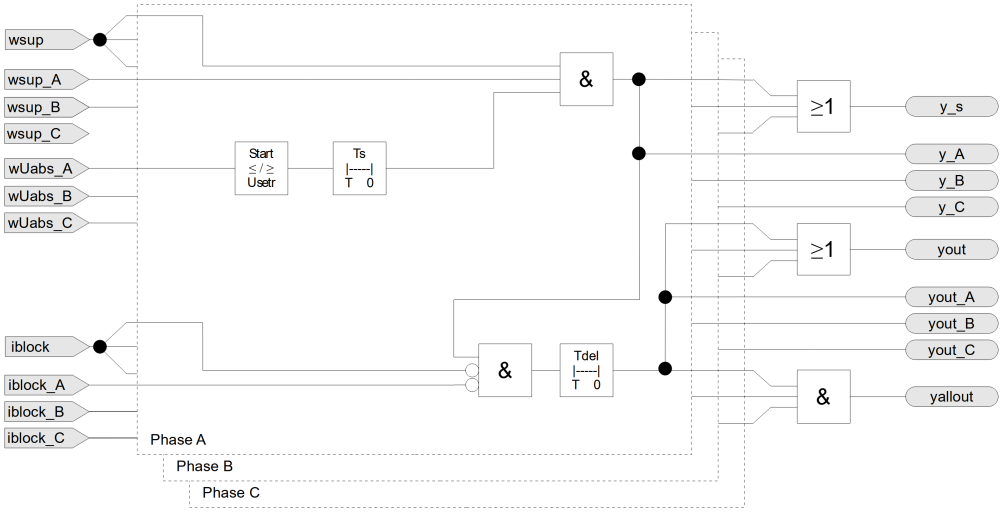


Figure 3.3: Logic scheme of 3-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations

The starting signals (*s:y_X*) and corresponding to tripping signals (*s:yout_X*) are calculated as:

$$\text{y_X} = \max(\text{wsup}, \text{wsup_X}, \text{Start_X})$$

$$\text{yout_X} = \begin{cases} (\text{y_X} + \text{Tdel}), & \text{for} \quad (\text{y_n} + \text{Tdel}) < \min(\text{iblock}, \text{iblock_X}) \\ 9999.999, & \text{for} \quad (\text{y_X} + \text{Tdel}) \geq \min(\text{iblock}, \text{iblock_X}) \end{cases} \tag{7}$$

Where:

Tdel is the Time Delay in [s].

The phase combination signals are calculated as:

$$\text{y_s} = \min(\text{y_A}, \text{y_B}, \text{y_C})$$

$$\text{yout} = \min(\text{yout_A}, \text{yout_B}, \text{yout_C}) \tag{8}$$

$$\text{yallout} = \max(\text{yout_A}, \text{yout_B}, \text{yout_C})$$

Table 3.5: Input/output signals of the 3-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations

Signal	Description	Unit	Type
wsup	Supervising Signal	s	IN
wsup_A	Supervising Signal A	s	IN
wsup_B	Supervising Signal B	s	IN
wsup_C	Supervising Signal C	s	IN
wUabs_A	Voltage, Magnitude A	sec.V	IN
wUabs_B	Voltage, Magnitude B	sec.V	IN
wUabs_C	Voltage, Magnitude C	sec.V	IN

Table 3.5: Input/output signals of the 3-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations

Signal	Description	Unit	Type
iblock	Blocking Input	s	IN
iblock_A	Blocking Input A	s	IN
iblock_B	Blocking Input B	s	IN
iblock_C	Blocking Input C	s	IN
y_s	Starting (OR)	s	OUT
y_A	Starting A	s	OUT
y_B	Starting B	s	OUT
y_C	Starting C	s	OUT
yout	Tripping Time (OR)	s	OUT
yout_A	Tripping Time A	s	OUT
yout_B	Tripping Time B	s	OUT
yout_C	Tripping Time C	s	OUT
yallout	Tripping Time (AND)	s	OUT

Table 3.6: Parameters of the 3-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations

Parameter	Description	Unit
uabs:A	Voltage, Magnitude A	p.u.
uabs:B	Voltage, Magnitude B	p.u.
uabs:C	Voltage, Magnitude C	p.u.
Uabs:A	Voltage, Magnitude A	sec.V
Uabs:B	Voltage, Magnitude B	sec.V
Uabs:C	Voltage, Magnitude C	sec.V
Upabs:A	Voltage, Magnitude A	pri.V
Upabs:B	Voltage, Magnitude B	pri.V
Upabs:C	Voltage, Magnitude C	pri.V

3.2.2 RMS and EMT Simulation

The Figure 3.4 shows the structure of the 3-Phase Under-/Overvoltage Model for RMS and EMT Simulations.

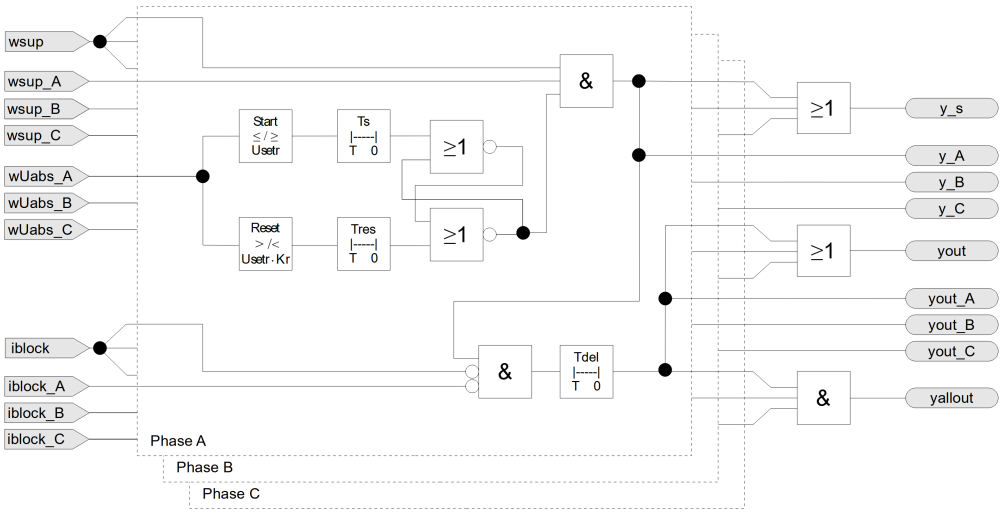


Figure 3.4: Logic scheme of the 3-Phase Under-/Overvoltage Model for RMS and EMT Simulations

The starting signals (*s:y_X*) and tripping signals (*s:yout_X*) is calculated for every Phase *X* is as:

$$\begin{aligned} \text{y_X} &= \text{wsup} \wedge \text{wsup_X} \wedge \neg(\neg\text{Start_X} \vee \text{Reset_X}) \\ \text{yout_X} &= \text{y_X} \wedge \neg\text{iblock} \wedge \neg\text{iblock_X} \end{aligned} \tag{9}$$

Table 3.7: Input/output signals of the 3-Phase Under-/Overvoltage Model for RMS and EMT Simulations

Signal	Description	Unit	Type
wsup	Supervising Signal		IN
wsup_A	Supervising Signal A		IN
wsup_B	Supervising Signal B		IN
wsup_C	Supervising Signal C		IN
wUabs_A	Line Voltage, Magnitude A	sec.V	IN
wUabs_B	Line Voltage, Magnitude B	sec.V	IN
wUabs_C	Line Voltage, Magnitude C	sec.V	IN
iblock	Blocking Input		IN
iblock_A	Blocking Input A		IN
iblock_B	Blocking Input B		IN
iblock_C	Blocking Input C		IN
y_s	Starting (OR)		OUT
y_A	Starting A		OUT
y_B	Starting B		OUT
y_C	Starting C		OUT
yout	Tripping (OR)		OUT
yout_A	Tripping A		OUT
yout_B	Tripping B		OUT
yout_C	Tripping C		OUT
yallout	Tripping (AND)		OUT

Table 3.8: Parameters of the 3-Phase Under-/Overvoltage Model for RMS and EMT Simulations

Parameter	Description	Unit
Ttrip	Tripping Time	s
Ttripall	Tripping Time (AND)	s
Ttripp:A	Tripping Time A	s
Ttripp:B	Tripping Time B	s
Ttripp:C	Tripping Time C	s
uabs:A	Voltage, Magnitude A	p.u.
uabs:B	Voltage, Magnitude B	p.u.
uabs:C	Voltage, Magnitude C	p.u.
Uabs:A	Voltage, Magnitude A	sec.V
Uabs:B	Voltage, Magnitude B	sec.V
Uabs:C	Voltage, Magnitude C	sec.V
Upabs:A	Voltage, Magnitude A	pri.V
Upabs:B	Voltage, Magnitude B	pri.V
Upabs:C	Voltage, Magnitude C	pri.V

3.3 6-Phase Under-/Overvoltage Model

The 6-Phase Under-/Overvoltage Model is used for the Type “Voltage (6ph)”. Functionally the model is identical to the 3-Phase Under-/Overvoltage Model, but depending on the *usedLoop* setting, either the line-ground or line-line voltages are used as voltage inputs.

3.3.1 Load Flow and Short-Circuit Calculation

The Figure 3.5 shows the structure of the 6-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations.

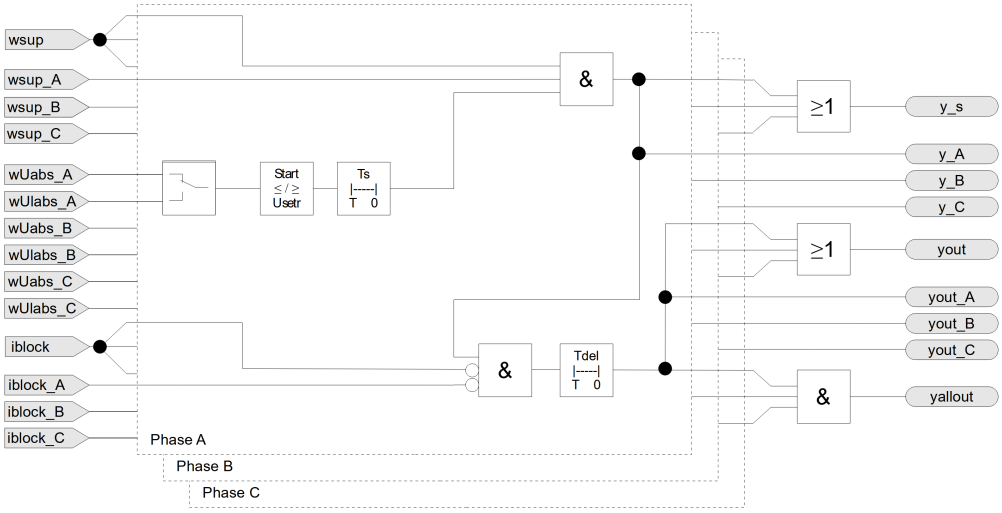


Figure 3.5: Logic scheme of 6-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations

Table 3.9: Input/output signals of the 6-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations

Signal	Description	Unit	Type
wsup	Supervising Signal	s	IN
wsup_A	Supervising Signal A	s	IN
wsup_B	Supervising Signal B	s	IN
wsup_C	Supervising Signal C	s	IN
wUabs_A	Line-Ground Voltage, Magnitude A	sec.V	IN
wUlabs_A	Line-Line Voltage, Magnitude A	sec.V	IN
wUabs_B	Line-Ground Voltage, Magnitude B	sec.V	IN
wUlabs_B	Line-Line Voltage, Magnitude B	sec.V	IN
wUabs_C	Line-Ground Voltage, Magnitude C	sec.V	IN
wUlabs_C	Line-Line Voltage, Magnitude C	sec.V	IN
iblock	Blocking Input	s	IN
iblock_A	Blocking Input A	s	IN
iblock_B	Blocking Input B	s	IN
iblock_C	Blocking Input C	s	IN
y_s	Starting (OR)	s	OUT
y_A	Starting A	s	OUT
y_B	Starting B	s	OUT
y_C	Starting C	s	OUT
yout	Tripping Time (OR)	s	OUT
yout_A	Tripping Time A	s	OUT
yout_B	Tripping Time B	s	OUT
yout_C	Tripping Time C	s	OUT
yallout	Tripping Time (AND)	s	OUT

Table 3.10: Parameters of the 6-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations

Parameter	Description	Unit
uabs:A	Voltage, Magnitude A	p.u.
uabs:B	Voltage, Magnitude B	p.u.
uabs:C	Voltage, Magnitude C	p.u.
Uabs:A	Voltage, Magnitude A	sec.V
Uabs:B	Voltage, Magnitude B	sec.V
Uabs:C	Voltage, Magnitude C	sec.V
Upabs:A	Voltage, Magnitude A	pri.V
Upabs:B	Voltage, Magnitude B	pri.V

Table 3.10: Parameters of the 6-Phase Under-/Overvoltage Model for use in Load Flow and Short-Circuit Calculations

Parameter	Description	Unit
Upabs:C	Voltage, Magnitude C	pri.V

3.3.2 RMS and EMT Simulation

The Figure 3.6 shows the structure of the 6-Phase Under-/Overvoltage Model for RMS and EMT Simulations.

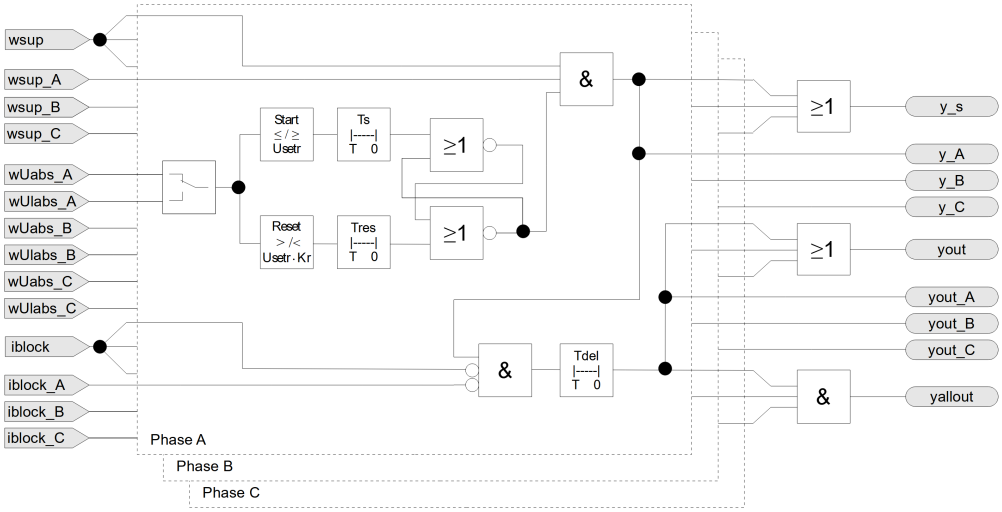


Figure 3.6: Logic scheme of the 6-Phase Under-/Overvoltage Model for RMS and EMT Simulations

Table 3.11: Input/output signals of the 6-Phase Under-/Overvoltage Model for RMS and EMT Simulations

Signal	Description	Unit	Type
wsup	Supervising Signal		IN
wsup_A	Supervising Signal A		IN
wsup_B	Supervising Signal B		IN
wsup_C	Supervising Signal C		IN
wUabs_A	Line-Ground Voltage, Magnitude A	sec.V	IN
wUlabs_A	Line-Line Voltage, Magnitude A	sec.V	IN
wUabs_B	Line-Ground Voltage, Magnitude B	sec.V	IN
wUlabs_B	Line-Line Voltage, Magnitude B	sec.V	IN
wUabs_C	Line-Ground Voltage, Magnitude C	sec.V	IN
wUlabs_C	Line-Line Voltage, Magnitude C	sec.V	IN
iblock	Blocking Input		IN
iblock_A	Blocking Input A		IN
iblock_B	Blocking Input B		IN
iblock_C	Blocking Input C		IN
y_s	Starting (OR)		OUT
y_A	Starting A		OUT
y_B	Starting B		OUT
y_C	Starting C		OUT
yout	Tripping (OR)		OUT
yout_A	Tripping A		OUT
yout_B	Tripping B		OUT
yout_C	Tripping C		OUT
yallout	Tripping (AND)		OUT

Table 3.12: Parameters of the 6-Phase Under-/Overvoltage Model for RMS and EMT Simulations

Parameter	Description	Unit
Ttrip	Tripping Time	s
Ttripall	Tripping Time (AND)	s
Ttripp:A	Tripping Time A	s
Ttripp:B	Tripping Time B	s
Ttripp:C	Tripping Time C	s
uabs:A	Voltage, Magnitude A	p.u.
uabs:B	Voltage, Magnitude B	p.u.
uabs:C	Voltage, Magnitude C	p.u.
Uabs:A	Voltage, Magnitude A	sec.V
Uabs:B	Voltage, Magnitude B	sec.V
Uabs:C	Voltage, Magnitude C	sec.V
Upabs:A	Voltage, Magnitude A	pri.V
Upabs:B	Voltage, Magnitude B	pri.V
Upabs:C	Voltage, Magnitude C	pri.V

3.4 3-Phase Vector-Shift Model

When in the Function *ifunc* combo box the Vector shift item has been set, the block monitors the angle of the phase voltages and trips when the variation of the angle of phases is greater than the Delta Angle *rUang* parameter value. This model has been developed to be used running an EMT simulation, but it can be also used running a RMS simulation or a SHC calculation. The algorithm is also slightly different in the static calculation (short circuit calculation) to the dynamic simulation (RMS and EMT).

3.4.1 Static calculation

In the short circuit calculation, except the angular differences calculated at the short circuit time, the only other data available are the load flow data. For this reason the angle variation is calculated as the difference between the angles calculated running the load flow and the angles calculated running the short circuit calculation. If the angle variation of all phases is greater than the Delta Angle *rUang* parameter value then a Vector shift is detected and the block trips

3.4.2 Dynamic simulation

The RMS/EMT algorithm calculates the angular difference between the actual voltage angles and the angles measured a cycle and an half cycle before the actual simulation time. If at least 5 of these 6 angular differences are greater than the Delta Angle *rUang* parameter value then a Vector shift is detected and the block trips.

Table 3.13: Input/output signals of 3 phase Vector Shift element (*CalUvecshift*)

Signal	Description	Unit	Type
wsup	Supervising Signal		IN
wsup_A	Supervising Signal A		IN
wsup_B	Supervising Signal B		IN
wsup_C	Supervising Signal C		IN
wUr_A	Phase A voltage real part	sec.V	IN
wUi_A	Phase A voltage imaginary part	sec.V	IN
wUr_B	Phase B voltage real part	sec.V	IN
wUi_B	Phase B voltage imaginary part	sec.V	IN
wUr_C	Phase C voltage real part	sec.V	IN
wUi_C	Phase C voltage imaginary part	sec.V	IN
clock	Clock signal		IN
iblock	Blocking Input		IN
iblock_A	Blocking Input A		IN
iblock_B	Blocking Input B		IN
iblock_C	Blocking Input C		IN
y_s	Starting (OR)		OUT
y_A	Starting A		OUT
y_B	Starting B		OUT
y_C	Starting C		OUT
yout_A	Tripping A		OUT
yout_B	Tripping B		OUT
yout_C	Tripping C		OUT
yallout	Tripping (AND)		OUT

Table 3.14: Parameters of 3 phase Vector Shift element (*CalUvecshift*)

Parameter	Description	Unit
Ttripall	Tripping Time (AND)	s
Ttripp:A	Tripping Time A	s
Ttripp:B	Tripping Time B	s

Table 3.14: Parameters of 3 phase Vector Shift element (*CalUvecshift*)

Parameter	Description	Unit
Ttripp:C	Tripping Time C	s

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